Lifelong Cognitive–Cyber Digital Twin

Notification Audio Monitor — LCDT Framework Mapping

Workflow Problem Solved Using Official LCDT Symbols & Formulas

1. Architectural Overview

The Cyber Watchdog implements a 5-layer Digital Twin architecture that models the human internal state as a dynamical system. This document outlines the mathematical mapping from raw sensor observations to predictive risk outcomes.

Layer 1 — Human Sensing: Multimodal Observation Space X_t

Every 10 seconds the system gathers the raw signal vector. The LCDT notation defines this as the Observation X_t containing 12 parameters across four signal families:

Signal Family	Parameter in X_t	Our Monitor's Variable	Status
Adversarial Exposure	Phishing simulations, Scam tasks	peak > 0.05 threshold	✓ Mapped
Digital Behaviour	Password habits, Browser exposure	target_processes list	✓ Mapped
Work Context	Notifications	notification_count	✓ Mapped
Cognitive Biomarkers	Task switching, Typing errors	Not in this module	– Partial

Layer 2 — Latent Cognitive–Cyber State Inference

Layer 2 uses a Bayesian Deep State-Space Model to map the raw observations X_t into the four Hidden Roots of the internal state vector:

$$Z_t = [\ C_t\ ,\ D_t\ ,\ H_t\ ,\ A_t\ ]$$

Symbol	Name	What it Means	Driven by	Range
C_t	**Cognitive Capacity (Cyan)**	Mental energy available — reduced by fatigue & sleep deficit	Vision Fatigue, Sleep Deficit, Late-night work	0.0 → 1.0
D_t	**Cognitive Demand (Magenta)**	Stress load — raised by interruptions & task-switching	Task Switches, Notifications ← our sensor	0.1 → 1.0
H_t	**Habit State (Green)**	Quality of learned security behaviour	Password manager use, Browser exposure	0.0 → 1.0
A_t	**Adversarial Exposure (Red)**	Active threat level — spikes on phishing events	Phishing simulations, Scam task interactions	0.0 → 1.0

The output of Layer 2 is the posterior distribution $\mathbf{P(Z_t \mid X_{1:t})}$ — the Digital Twin's current internal state given all observations up to time t. Our notification count feeds directly into D_t (Demand) via the Work Pressure sub-calculation.

Layer 3 — Multi-Timescale Dynamical System (Core Novelty)

Layer 3 gives each hidden state momentum using the Euler Method over a 10-second interval Δt. Three timescales are modelled simultaneously:

3a. Slow Dynamics — Cognitive Capacity C_t (Years timescale)

$dC_t = \alpha (\mu - C_t) dt + \sigma dW_t$

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α (Recovery Speed)	**0.01**	very slow; capacity takes hours to recharge
μ (Target Baseline)	**1.0**	normally; drops toward 0 when sleep deficit is detected in X_t
σdW_t	**—**	Stochastic noise term (Wiener process) — models biological variability
C_t in our code	**1.0**	Assumed full energy in the worked example; no fatigue sensor yet

3b. Fast Dynamics — Cognitive Demand D_t (Hours timescale)

$dD_t = [f(work) - \lambda (D_t - 0.1)] dt + \sigma dW_t$

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f(work)	**—**	Work Pressure — computed from Task Switches + Notifications count in X_t
λ (Cooling Rate)	**0.2**	stress decays over a few minutes of inactivity
0.1	**—**	Resting baseline; demand never drops below idle level
Notifications role	**—**	Each audio peak detected by our sensor increments f(work), raising D_t

3c. Medium Dynamics — Habit State H_t (Weeks timescale)

$H_{t+1} = H_t + \eta \cdot R_t$

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η (Learning Rate)	**0.1**	habit changes gradually over weeks
R_t (Reward)	**+0.05**	for good security behaviour (password manager used)
R_t (Penalty)	**–0.10**	for bad security behaviour (manual password typed, HTTP site)
H_t in example	**0.8**	user has mostly good habits

3d. Cognitive Reserve Gap — CRG_t (The Core Logic)

$$CRG_t = C_t - D_t$$

Positive CRG_t	Spare brainpower available → Healthy, lower risk
Negative CRG_t	Cognitive debt → Risky, higher mistake probability
Note (PDF sign convention)	*The PDF defines $CRG_t = D_t - C_t$ in Layer 4 formula; the β_1 weight is negative (–3) to flip direction correctly*

Layer 4 — Cyber Risk Forecasting Engine

Layer 4 converts the hidden state vector Z_t into a single Mistake Probability $P(M_t)$ using a logistic regression over the three key roots.

Step 1 — Logit Score (Linear Combination with Beta Weights):

$Score = \beta_1 \cdot CRG_t + \beta_2 \cdot H_t + \beta_3 \cdot A_t + bias$	
$Score = (-3 \times CRG_t) + (-3 \times H_t) + (6 \times A_t) + 0.5$	
$\beta_1 = -3$	Reserve Gap weight — higher spare capacity lowers risk (negative sign)
$\beta_2 = -3$	Habit Quality weight — better habits lower risk (negative sign)
$\beta_3 = +6$	Adversarial Threat weight — active attack massively raises risk (double weight)
Bias = +0.5	Baseline offset — slight positive risk even in ideal conditions

Step 2 — Sigmoid Squash to Probability:

$$P(M_t) = \sigma(\ Score\ ) = 1 / (\ 1 + e^{-Score}\ )$$

Worked Example — High-Load Simulation Scenario

We solve a high-stress cognitive scenario using the LCDT symbols and formulas. This example demonstrates how multi-modal pressure translates into a concrete risk score.

High-Stress Scenario (Active Task Switching & Phishing Interaction)		
Layer 1 — X_t[Task Switches]	**20**	user jumping between windows
Layer 1 — X_t[Notifications]	**10**	captured from audio sensors
Layer 1 — X_t[Phishing]	**1**	user clicked unknown email link
f(work)	**0.5**	normalized from 20 task switches + 10 notification peaks
Initial States	**1.0, 0.1, 0.8, 0.0**	Capacity, Demand, Habits, Adversarial
New Demand (D_{t+1})	**0.6**	0.1 + [0.5 – 0.2(0.1–0.1)] = 0.6
Adversarial (A_{t+1})	**1.0**	phishing click instant spike
Habits (H_{t+1})	**0.79**	0.8 + 0.1 × (–0.1) — slight penalty for phishing click
Reserve Gap (CRG_t)	**0.4**	1.0 – 0.6 = 0.4
Score	**2.9**	(–3×0.4)+(–3×0.8)+(6×1.0)+0.5 = –1.2–2.4+6.0+0.5

FINAL RESULT: $P(M_t)$ = 94.7% Mistake Risk

Layer 5 — Digital Twin Simulation Engine: $E[\ M_{t:T} \mid do(\pi)\ ]$

Layer 5 is the counterfactual brain of the Digital Twin. The notation **do(π)** represents a forced Policy Intervention simulated over 30 days to answer 'What If?' questions.

Intervention do(π)	What We Force	Effect on Z_t	Expected $P(M_t)$
do(UX Optimisation)	Cap X_t[Notifications]=0, Task Switches=0	D_t stays near 0.1 (resting); CRG_t→0.9 (very healthy)	**↓ drops significantly**

do(Security Training)	Force daily $\eta \cdot R_t$ spike in H_t formula	$H_t \to 1.0$ over ~30 days; $\beta_2 \cdot H_t$ term maximally protective	↓ **moderate drop**
do(Increased Workload)	Force f(work)=1.0 (max) for all 30 days	D_t rises; CRG_t goes negative → Cognitive Debt	↑ **risk escalates**
do(Aging Progression)	Decrease μ (Capacity Baseline) by 2% per simulated day	C_t ceiling drops slowly; CRG_t shrinks even at same workload	↑ **gradual rise**
Baseline (no intervention)	Current behaviour continues unchanged	If already at cognitive debt, line climbs as C_t drains	→ **natural trajectory**

Complete Symbol Glossary

Symbol	Full Name	Meaning in This Workflow
X_t	**Observation vector at time t**	All 12 raw sensor readings every 10 seconds, including notification audio peaks
Z_t	**Latent hidden state**	The four internal cognitive-cyber roots inferred from X_t
C_t	**Cognitive Capacity**	Mental energy tank (cyan). Starts at 1.0, drains slowly with fatigue/age
D_t	**Cognitive Demand**	Stress load (magenta). Raised by notifications and task-switching
H_t	**Habit State**	Security behaviour quality (green). Updated by $\eta \cdot R_t$ each interval
A_t	**Adversarial Exposure**	Threat level (red). Spikes instantly to 1.0 on phishing click
CRG_t	**Cognitive Reserve Gap**	$C_t - D_t$. Positive = healthy buffer. Negative = cognitive debt
$P(M_t)$	**Mistake Probability**	Final risk output, 0–100%. $\sigma(\beta_1 \cdot CRG + \beta_2 \cdot H + \beta_3 \cdot A)$
$do(\pi)$	**Policy Intervention**	A forced change applied in Layer 5 counterfactual simulation
α	**Recovery Speed**	0.01 — rate at which C_t returns to baseline μ
μ	**Target Capacity Baseline**	1.0 normally; reduced when sleep deficit detected in X_t
λ	**Cooling Rate**	0.2 — rate at which D_t decays back to resting level 0.1
η	**Learning Rate**	0.1 — how quickly H_t responds to positive/negative behaviour
R_t	**Behaviour Reward**	+0.05 for good security; –0.10 for bad security actions
$\beta_{1,2,3}$	**Beta Weights**	Importance factors: β_1=–3 (CRG), β_2=–3 (H), β_3=+6 (A)
$\sigma(\cdot)$	**Sigmoid function**	$1/(1+e^{-x})$; squashes any score into 0–1 probability
dW_t	**Wiener Process**	Stochastic noise term modelling biological/behavioural randomness

Summary: The Digital Twin Mapping

The Cyber Watchdog translates raw sensor signals into a predictive Digital Twin using 5 mathematical layers. By capturing activity like task-switching and notifications, the system builds a live model of Cognitive Demand (D_t) and Capacity (C_t). This allows the system to accurately forecast the Mistake Probability $P(M_t)$ and simulate long-term security outcomes using the Layer 5 engine, ensuring a proactive and privacy-first approach to cybersecurity.

Full 5-Layer Algorithm Walkthrough: "Late-Night Crunch"

Layer 1 — Observation (X_t)

Input FAMILY	`Signals Captured`	*Reasoning (Why)*
Work Context	`Task Switches = 25`	*High volume context switching increases Demand floor.*
Distractions	`Notifications = 12`	*Audio pings directly increment the work pressure function.*
Biomarkers	`Time = 23:45 (Night)`	*Biological clock signals a drop in Energy baseline (μ).*
Threat Familien	`Phishing = 1.0 (Click)`	*Adversarial family triggers max threat spike.*

Layer 2 — State Inference (Z_t)

Calculation	`Result`	*Logic (What)*
Work Pressure	`f(work) = 0.65`	*Normalized switches (25/40) + notifications (12/10) capped.*
Energy Floor	$\mu = 1.0 - 0.2 = 0.8$	*Night-shift penalty applied to standard 1.0 baseline.*
Habit Reward	$R_t = -0.1$	*Penalty assigned for clicking a suspicious link.*
Inference P(Z)	`Z = [C, D, H, A]`	*Mapping raw X_t into the 4 latent hidden roots.*

Layer 3 — Dynamics (Euler Step Equations)

Formula / Step	`Arithmetic Solving`	*Dynamic Effect*
dD_t Eq.	$(0.65 - 0.2 \times (0.1 - 0.1)) \times 1.0$	*Demand increases from 0.1 to 0.75 instantly.*
dC_t Eq.	$0.01 \times (0.8 - 1.0) \times 1.0$	*Capacity drains from 1.0 to 0.98 due to fatigue.*
H_{t+1} Eq.	$0.8 + 0.1 \times (-0.1)$	*Habits drop from 0.8 to 0.79 (Protective Shield weak).*
CRG_t Gap	$C_t - D_t = 0.98 - 0.75$	*Positive buffer is only 0.23 (Critical level).*

Layer 4 — Risk Score (Logistic Regression)

Weight Mapping	**Step-by-Step Summation**	*Statistical Meaning*
β■ · CRG	$-3 \times 0.23 = -0.69$	*Reserve Gap impact (Low reserve = More risk).*
β■ · H	$-3 \times 0.79 = -2.37$	*Habit impact (Weak habits = More risk).*
β■ · A	$6 \times 1.0 = 6.0$	*Threat impact (Active attack = Max risk).*
Final Logit	$-0.69 - 2.37 + 6.0 + 0.5$	*Logit Score = 3.44 (Severe danger zone).*
$P(M_t)$	$1 / (1 + e^{-3.44})$	*Result: 96.88% probability of a critical mistake.*

Layer 5 — Simulation (Counterfactual do-calculus)

What-If Scenario	**Formula Change**	*Predicted Impact*
do(UX Optim)	Force X_t[Alerts] = 0	*Eliminating interruptions to preserve the gap.*
New dD_t	$(0.25 - 0.2 \times 0) \times 1.0$	*New demand would be 0.35 instead of 0.75.*
New Score	$-3(0.63) - 3(0.79) + 6(1) + 0.5$	*New Score = 2.24 (Medium danger zone).*
Simulation Result	$P(M_t) \to 90.3\%$	*Even under attack, UX fix saves ~6% risk immediately.*